

SANYAM KAUL

kaulsanyam@gmail.com — +91-7018498338

EDUCATION

M.Tech Computer Science & Engineering

IIT Hyderabad

2025 CGPA: 8.90

B.Tech Computer Science & Engineering (Networks & Security)

VIT-AP University, Amaravati

2021 CGPA: 8.90

EXPERIENCE

Data Engineer

Netradyne

July 2025 - Present

- Implemented AWS Monitoring Dashboard enabling users to view EC2 instances with metrics and cost analysis, reducing infrastructure costs
- Enhanced Kubernetes EKS logging infrastructure by integrating rsyslog sidecar containers for real-time log monitoring and forwarding
- Developed frontend functionality for log download using Elasticsearch integration, improving debugging efficiency for CI/CD pipelines
- Optimized TensorRT-based reprocessing pipelines (v7, v8, v10) for analytics code compilation and Docker image management
- Enhanced alert-kpi tool with _ALL locale support for dataset creation, enabling model training teams to train on aggregated datasets

Tech Analyst (SDE-1)

BlackRock

July 2021 - June 2023

- Developed full-stack Self-Service Financial Information Reporting Tool (Angular, Spring Boot), reducing report setup time from 1.5 months to 2-3 days
- Engineered scalable applications using Angular, Python, and Spring Boot with 85%+ test coverage, focusing on architecture and design patterns
- Implemented CI/CD pipelines on Azure DevOps for 5 business-critical servers, achieving 50% automation in performance and integration testing
- Spearheaded REST and GraphQL API integration, standardizing workflows across 50+ business, development, and QA teams
- Mentored 3 interns through BlackRock's Buddy program, facilitating smooth onboarding and technical skill development

Tech Intern

BlackRock

January 2021 - July 2021

- Led team of 4 to build Angular and Golang Dashboard for visualizing CI/CD pipeline failures with Azure DevOps integration
- Built real-time backend service using Golang with Azure APIs to fetch and process CI/CD metrics and failure statistics

PROJECTS

WineSecretHK - RAG-based AI Food & Wine Pairing System

rag.winesecrethk.com/docs

- Built RAG-based AI system with HyDE-inspired algorithm for flavor profile matching using cosine similarity on vector embeddings; algorithm submitted for patent
- Implemented multi-modal input processing (text, audio, images) with automatic query intent identification using FastAPI backend, AWS RDS, and ChromaDB
- Integrated Stripe payments and Eats365 ordering platform via webhooks and server-side callbacks

IITH FoML Hackathon 2023 - 1st Place

IIT Hyderabad

2023

- Won 1st place with voting-based ensemble model combining Random Forest and Gradient Boosting classifiers
- Implemented advanced ensemble learning techniques to improve prediction accuracy through model combination

IITH Deep Learning Hackathon 2024 - 15th/100+ Teams

IIT Hyderabad

2024

- Ranked 15th out of 100+ teams by implementing and comparing ResNet variants, DenseNet, ResNext, and Vision Transformer (ViT)
- Performed extensive hyperparameter tuning and model comparison to achieve competitive results on image classification tasks

- Engineered complex architectures from scratch: CNN, RNN, Vision Transformer, Autoencoder, Variational Autoencoder using NumPy
- Implemented core algorithms including backpropagation, MLP, and optimization techniques, gaining deep understanding of neural network internals

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, Golang, C++
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **ML Algorithms:** Random Forest, Gradient Boosting, Ensemble Methods, CNN, RNN, ViT, Autoencoders, PCA, K-means, K-NN
- **Frameworks:** Spring Boot, Angular, Express.js, FastAPI
- **Cloud & DevOps:** AWS (EC2, EKS, S3, FSx, CloudWatch, RDS), Azure DevOps, Docker, Kubernetes, Jenkins, CI/CD
- **Databases:** MySQL, MongoDB, Sybase, ChromaDB, AWS RDS
- **AI/ML Concepts:** RAG, Vector Embeddings, Backpropagation, Gradient Descent, Hyperparameter Tuning, Model Evaluation
- **Tools:** Git, Jira, Confluence, Elasticsearch, REST APIs, GraphQL, Jupyter Notebooks

ACHIEVEMENTS

- 4/5 star rating in BlackRock annual employee ratings (2021, 2022); Promotion nomination to Associate (SDE-2) for 2024
- 1st position in IITH ML Hackathon 2023; 15th position (out of 100+ teams) in IITH Deep Learning Hackathon 2024
- 1st position in Object Oriented Programming course (100+ students); 450+ problems solved on LeetCode